

End-to-End Supply Chain Network & Logistics Optimization

Motupalli Hemanth Kumar

Abstract

This report outlines the formulation and resolution of a Mixed-Integer Linear Programming (MILP) model designed to optimize a multi-echelon supply chain network. By modeling capacitated network flows and dynamic facility allocation, the system efficiently satisfies distribution demand across varying regional nodes while minimizing overall logistical costs.

1 Problem Description

A petroleum distribution network manages the flow of resources from major supply ports to central demand centers via a series of intermediate staging depots. The primary objective is to determine the optimal continuous flow quantity through each transportation link (pipeline) to minimize the total operational cost, ensuring all regional demands are met without accumulating local residual storage.

2 Mathematical Formulation

Let N denote the set of all facilities (nodes) and E denote the set of directed logistical edges (i, j) representing the pipelines.

2.1 Parameters

- S_n : The net supply at node $n \in N$ (positive for supply generation, negative for required demand, and zero for pure transshipment nodes).
- c_{ij} : The unit transportation and processing cost along edge $(i, j) \in E$.

2.2 Decision Variables

- $x_{ij} \geq 0$: The continuous quantity of product transported through edge $(i, j) \in E$.

2.3 Objective Function

The objective is to minimize the total transportation cost across the supply chain:

$$\min \sum_{(i,j) \in E} c_{ij} x_{ij}$$

2.4 Constraints

Flow Conservation: The net logistical flow at each node must perfectly balance its required supply or demand properties.

$$\sum_{j:(i,j) \in E} x_{ij} - \sum_{k:(k,i) \in E} x_{ki} = S_i \quad \forall i \in N$$

3 Scenario Analysis and Results

The optimization models were formulated utilizing the Pyomo modeling framework in Python and optimally solved via the CBC (COIN-OR Branch and Cut) solver algorithm.

3.1 Base Optimization Case

The optimal uncapacitated network flow satisfied all distribution requirements with a minimized total distribution cost of **2500 units**. The algorithm’s flow distribution successfully prioritized high-capacity, low-cost pipelines, notably routing 190 total units directly through the Solapur-Gulbarga distribution link.

3.2 Capacitated and Disrupted Network

To rigorously test the resilience and dynamic allocation of the network, a structural disruption was introduced where the primary Solapur-Gulbarga pipeline was completely shut down. Concurrently, a strict capacity limit was placed on the Bangalore depot, capping its maximum processing inflow at 150 units.

The MILP solver successfully re-routed the network flow dynamically around the blockages, yielding a newly optimized operational cost of **2690 units**. The logistical flow through alternative hubs (Hubli and Mangalore) automatically scaled up to compensate for the disrupted path, all while strictly adhering to the newly introduced facility capacity constraints.

4 Conclusion

The formulated MILP effectively optimized the multi-echelon network, providing a highly scalable and mathematically rigorous framework for distribution planning. The subsequent integration of strict capacity limits and structural network disruptions firmly demonstrated the model’s ability to maintain feasibility and cost-efficiency under volatile logistical conditions.